Exam 1 Review

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① $1010\ 0101 \times 10 =$

$1010\ 0101_2 = 2^0+2^2+2^5+2^7 = 165_{10}$ $10_2 = 2_{10}$

$$\begin{array}{r} 1010\ 0101 \\ \times\ \ \cancel{10} \\ \hline 101001010 \end{array} \qquad \begin{array}{r} 165 \\ \times\ 2 \\ \hline 330 \end{array}$$

overflow

$(1)\ 0100\ 1010_2$ 330_{10}

$330_{10} > 256_{10}$; result must "wrap around"!

$\therefore\ 256\overline{)330_{10}}$, -256, 074 $330 \% 256 = \boxed{74_{10}}$ stored value

$\boxed{74_{10} = 0100\ 1010_2}$

②

(a) $1011\ 1000$ (<0)

$$\begin{array}{r} 1011\ \cancel{1000} \\ -\qquad\ \ 1 \\ \hline 1011\ 0111 \end{array}$$

$\downarrow$

$0100\ 1000_2$

$0100\ 1000_2 = 2^3+2^6 = 72_{10}$

Negative $\boxed{-72_{10}}$

(b) -85

$$\begin{array}{ll} 0 & R1 \\ 2\overline{)1} & R0 \\ 2\overline{)2} & R1 \\ 2\overline{)5} & R0 \\ 2\overline{)10} & R1 \\ 2\overline{)21} & R0 \\ 2\overline{)42} & R1 \\ 2\overline{)85} & \end{array}$$

$85_{10} = 1010101_2$

$= 0101\ 0101_2$

$\downarrow$

$1010\ 1010$
$+\qquad\quad 1$
$1010\ 1011$

$\boxed{-85_{10} = 1010\ 1011_2}$

③

(a) 1023_5

$1023_5 = (3\times5^0)+(2\times5^1)+(1\times5^3) = 3+10+125 = 138_{10}$

$$\begin{array}{ll} 0 & R2 \\ 8\overline{)2} & R1 \\ 8\overline{)17} & R2 \\ 8\overline{)138} & \\ -8\downarrow & \\ 58 & \\ -56 & \\ 2 & \end{array}$$

$138_{10} = \boxed{212_8}$

(b) 762_8

$762_8 = (2\times8^0)+(6\times8^1)+(7\times8^2) = 2+48+448 = 498_{10}$

$$\begin{array}{ll} 0 & R3 \\ 5\overline{)3} & R4 \\ 5\overline{)19} & R4 \\ 5\overline{)99} & R3 \\ 5\overline{)498} & \\ -45\downarrow & \\ 48 & \\ -45 & \\ 3 & \end{array}$$

$498_{10} = \boxed{3443_5}$

④

(a) 0111.1011_2

$2^2+2^1+2^0+2^{-1}+2^{-3}+2^{-4} = 4+2+1+\frac{1}{2}+\frac{1}{8}+\frac{1}{16} = \boxed{7.6875_{10}}$

(b) 10101.101_2

$2^4+2^2+2^0+2^{-1}+2^{-3} = 16+4+1+\frac{1}{2}+\frac{1}{8} = \boxed{21.625_{10}}$

⑤

(a) $f = -1.101_2 \times 2^{1100_2}$, Bias 7

$1100_2 = 2^3+2^2 = 12_{10} - 7 = 5_{10}$

$1.101_2 = 2^0+2^{-1}+2^{-3} = 1.625_{10}$

$-1.625\times2^5 = \boxed{-52_{10}}$

(b) $f = +1.010_2 \times 2^{0101_2}$, Bias 7

$0101_2 = 5_{10} - 7 = -2_{10}$

$1.010_2 = 2^0+2^{-2} = 1.25_{10}$

$1.25\times2^{-2} = \boxed{0.3125_{10}}$

⑥

Mult. 2 n×n matrices has complexity $\boxed{O(n^3)}$

vector × vector = $O(n)$

vector × matrix = $O(n^2)$

⑦ $\sqrt[4]{1+x} - \cos(x)$, 3 sig. digs., $x = 0.01$

(a) Exact result: $0.002540679 \approx \boxed{0.00254}$

(b) Reformulate

$\cos^2(x) = 1-\sin^2(x)$
$-\cos^2(x) = \sin^2(x)-1$

$$\sqrt[4]{1+x}-\cos(x)\cdot\frac{\sqrt[4]{1+x}+\cos(x)}{\sqrt[4]{1+x}+\cos(x)} = \frac{\sqrt{1+x}-\cos^2(x)}{\sqrt[4]{1+x}+\cos(x)}$$

$$= \frac{(\sqrt{1+x}-1)(\sqrt{1+x}+1)}{(\sqrt[4]{1+x}+\cos(x))(\sqrt{1+x}+1)} + \frac{\sin^2(x)}{\sqrt[4]{1+x}+\cos(x)} = \frac{\cancel{1}+x-\cancel{1}}{(\sqrt[4]{1+x}+\cos(x))(\sqrt{1+x}+1)} + \frac{\sin^2(x)}{\sqrt[4]{1+x}+\cos(x)} = \boxed{\frac{x}{(\sqrt[4]{1+x}+\cos(x))(\sqrt{1+x}+1)} + \frac{\sin^2(x)}{\sqrt[4]{1+x}+\cos(x)}}$$

(c) Original result

$\sqrt[4]{1+0.01} - \cos(0.01) = 1.00-1.00 = \boxed{0.00}$

Relative error: $\frac{|0.00254-0.00|}{0.00254}\times100\% = \boxed{100\%}$

(d) Reform. result

$$\frac{0.01}{(\sqrt[4]{1+0.01}+\cos(0.01))(\sqrt{1+0.01}+1)} + \frac{\sin^2(0.01)}{\sqrt[4]{1+0.01}+\cos(0.01)} = \frac{0.01}{(1.00+1.00)(1.00+1)} + \frac{\sin^2(0.01)}{1.00+1.00} = 0.0025+0.00005 = \boxed{0.00255}$$

Rel. err.: $\frac{|0.00254-0.00255|}{0.00254} = \boxed{0.394\%}$

⑧ $\sqrt{\pi+x}-\sqrt{\pi}$, $x = 0.01$, 3 sig. digs.

(a) $\boxed{0.00282}$

(b) $f(b)-f(a) = (b-a)f'(c)$

$f(b) = \sqrt{\pi+x}$ $f(a) = \sqrt{\pi} \rightarrow f(x) = \sqrt{x}$ $f'(x) = \frac{1}{2\sqrt{x}}$

$b = \pi+x$ $a = \pi$

$\sqrt{\pi+x}-\sqrt{\pi} = (\cancel{\pi}+x-\cancel{\pi})\cdot\frac{1}{2\sqrt{c}} = \frac{x}{2\sqrt{c}}$

$c = \frac{a+b}{2} = \frac{\pi+\pi+x}{2} = \pi+\frac{x}{2}$

$\sqrt{\pi+x}-\sqrt{\pi} = \frac{x}{2\sqrt{\pi+x/2}}$

(c) $\sqrt{\pi+0.01}-\sqrt{\pi} = 1.78-1.77 = \boxed{0.01}$

$\frac{|0.00282-0.01|}{0.00282}\times100\% = \boxed{255\%}$

(d) $\frac{0.01}{2\sqrt{\pi+\frac{0.01}{2}}} = \frac{0.01}{2\cdot1.77} = \boxed{0.00282}$ $\boxed{0\%}$

⑨ $\frac{1-\cos(x)}{x}$, $O(x^3)$, $x = 0.02$, 3 sig digs

(a) $\boxed{0.00100}$

(b) ~~$\cos(x) \approx 1-\frac{x^2}{2!}+\frac{x^4}{4!}-\frac{x^6}{6!}+\ldots$~~

~~$1-\cos(x) \approx \frac{x^2}{2!}-\frac{x^4}{4!}+\frac{x^6}{6!}+\ldots$~~

~~$\frac{1-\cos(x)}{x} \approx \frac{x}{2!}-\frac{x^3}{3!}+\ldots$~~

~~$\boxed{T(x) = \frac{x}{2}-\frac{x^3}{6}}$~~

$\cos(x) \approx \cos(0) - \cancel{\sin(0)x} - \frac{\cos(0)x^2}{2!} + \cancel{\frac{\sin(0)}{6!}x^3}$ ← cut off higher powers first

$= 1-\frac{x^2}{2!} = 1-\frac{x^2}{2}$

$1-\cos(x) \approx \frac{x^2}{2}$ $\frac{1-\cos(x)}{x} \approx \boxed{\frac{x}{2}}$

(c) $\frac{1-\cos(0.02)}{0.02} = \frac{1-1.00}{0.02} = \boxed{0.00}$ $\boxed{100\%}$

(d) $\frac{0.02}{2} = \boxed{0.01}$ $\boxed{0\%}$

⑮ $e^x \approx 1+x+\frac{x^2}{2!}+\frac{x^3}{3!}$

$e^{-x} \approx 1-x+\frac{x^2}{2!}-\frac{x^3}{3!}$

$\sin(x) \approx \cancel{\sin(0)}+\cos(0)x-\cancel{\frac{\sin(0)x^2}{2!}}-\frac{\cos(0)x^3}{3!}$

$= x-\frac{x^3}{6!}$

$\sin(x^2) = x^2-\cancel{\frac{x^4}{6!}}$

$f(x) = x^2-x^3$ rest cut off

$$\begin{array}{r|rrrr} 0.1 & -1 & 1 & 0 & 0 \\ & & -0.1 & 0.09 & 0.009 \\ \hline & -1 & 0.9 & 0.09 & 0.009 \end{array}$$

Rish practice Exam:

② Reformulate $1-\sqrt{\cos(x)}$ w/ trig

$$1-\sqrt{\cos(x)}\cdot\frac{1+\sqrt{\cos(x)}}{1+\sqrt{\cos(x)}} = \frac{1-\cos(x)}{1+\sqrt{\cos(x)}}\cdot\frac{1+\cos(x)}{1+\cos(x)} = \frac{1-\cos^2(x)}{(1+\sqrt{\cos(x)})(1+\cos(x))} = \boxed{\frac{\sin^2(x)}{(1+\sqrt{\cos(x)})(1+\cos(x))}}$$

③ Reform. $\frac{1}{x^2+2x+1}-\frac{1}{x^2}$ w/ MVT

$f(b)-f(a) = (b-a)f'(c)$

$f(b) = \frac{1}{x^2+2x+1}$

$f(a) = \frac{1}{x^2}$ $f(x) = \frac{1}{x^2}$ $f'(x) = -\frac{2}{x^3}$

$a = x$ $b = x+1$ $c = \frac{2x+1}{2}$

$$(\cancel{x}+1-\cancel{x})\left(-\frac{2}{\left(\frac{2x+1}{2}\right)^3}\right) = \boxed{-\frac{2}{\left(x+\frac{1}{2}\right)^3}}$$

④ Reform. $\cos(\pi+0.061)+1$

$\varepsilon = 0.061 \longrightarrow \cos(\pi+\varepsilon)+1$

$= \cos(\pi)\cos(\varepsilon)-\cancel{\sin(\pi)\sin(\varepsilon)}+1$

$= -\cancel{\cos(\pi)}\cos(\varepsilon)+1 = -\cos(\varepsilon)+1$ Mult. by conjugate

Error calc.

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⑤ $\sin(x)\cos(x)$ for $x = .1$; 5th-order $T(x)$ about 0

$f(x) = \sin(x)\cos(x)$ Product rule: $uv'+vu'$

$f'(x) = -\sin^2(x)+\cos^2(x)$

$f''(x) = -2\sin(x)\cos(x)-2\cos(x)\sin(x) = -4\sin(x)\cos(x)$

$f'''(x) = -4(\cos^2(x)-\sin^2(x))$

$f^{(4)}(x) = 16\sin(x)\cos(x)$

$f^{(5)}(x) = 16(\cos^2(x)-\sin^2(x))$

$f(0) = 0$ $f'(0) = 1$ $f''(0) = 0$

$f'''(0) = -4$ $f^{(4)}(0) = 0$ $f^{(5)}(0) = 16$

$p(x) = x-\frac{4}{3!}x^3+\frac{16}{5!}x^5 = \boxed{x-\frac{2}{3}x^3+\frac{2}{15}x^5}$

$$\begin{array}{r|rrrrrr} 0.1 & \frac{2}{15} & 0 & -\frac{2}{3} & 0 & 1 & 0 \\ & & 0.01333 & 0.001333 & -0.06653 & -0.006653 & 0.09933 \\ \hline & \frac{2}{15} & 0.01333 & -0.6653 & -0.06653 & 0.9933 & \boxed{0.09933} \end{array}$$